MSG No. 131.444 P_c4_2/mmc [Type IV, tetragonal]

Table 1: Wyckoff site: 2a, site symmetry: 4'/mmm'

No.	position	mapping
1	[0, 0, 0]	[1,4,5,6,9,12,13,14,18,19,23,24,26,27,31,32]
2	$\left[0, 0, \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16,17,20,21,22,25,28,29,30]

Table 2: Wyckoff site: 2b, site symmetry: 4'/m'mm'

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	[1,6,7,8,10,11,12,13,18,19,20,21,25,30,31,32]
2	$\left[0, 0, \frac{3}{4}\right]$	[2,3,4,5,9,14,15,16,17,22,23,24,26,27,28,29]

Table 3: Wyckoff site: 2c, site symmetry: 4'/mmm'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[1,4,5,6,9,12,13,14,18,19,23,24,26,27,31,32]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16,17,20,21,22,25,28,29,30]

Table 4: Wyckoff site: 2d, site symmetry: 4'/m'mm'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[1,6,7,8,10,11,12,13,18,19,20,21,25,30,31,32]
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[2,3,4,5,9,14,15,16,17,22,23,24,26,27,28,29]

Table 5: Wyckoff site: 4e, site symmetry: m'mm.

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[1,6,12,13,20,21,25,30]
2	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[2,3,15,16,23,24,26,27]
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[4,5,9,14,17,22,28,29]
4	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[7,8,10,11,18,19,31,32]

Table 6: Wyckoff site: 4f, site symmetry: mmm.

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	[1,4,5,6,9,12,13,14]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[2,3,7,8,10,11,15,16]
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[17,20,21,22,25,28,29,30]
4	$\left[\frac{1}{2}, 0, 0\right]$	[18,19,23,24,26,27,31,32]

Table 7: Wyckoff site: `4g`, site symmetry: `4'mm'`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,6,12,13,18,19,31,32]`
2	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[2,3,15,16,17,22,28,29]`
3	$[0,\ 0,\ -z]$	`[4,5,9,14,23,24,26,27]`
4	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[7,8,10,11,20,21,25,30]`

Table 8: Wyckoff site: `4h`, site symmetry: `4'mm'`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	`[1,6,12,13,18,19,31,32]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[2,3,15,16,17,22,28,29]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	`[4,5,9,14,23,24,26,27]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[7,8,10,11,20,21,25,30]`

Table 9: Wyckoff site: `8i`, site symmetry: `2mm.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,6,12,13]`
2	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	`[2,3,15,16]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[4,5,9,14]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	`[7,8,10,11]`
5	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[17,22,28,29]`
6	$\left[\frac{1}{2},\ 0,\ z\right]$	`[18,19,31,32]`
7	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[20,21,25,30]`
8	$\left[\frac{1}{2},\ 0,\ -z\right]$	`[23,24,26,27]`

Table 10: Wyckoff site: `8j`, site symmetry: `m.2'm'`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,14,23,32]`
2	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[2,11,21,28]`
3	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[3,10,20,29]`
4	$[x,\ -x,\ 0]$	`[4,13,19,26]`
5	$[-x,\ x,\ 0]$	`[5,12,18,27]`
6	$[-x,\ -x,\ 0]$	`[6,9,24,31]`
7	$\left[x,\ x,\ \frac{1}{2}\right]$	`[7,16,17,30]`
8	$\left[-x,\ -x,\ \frac{1}{2}\right]$	`[8,15,22,25]`

Table 11: Wyckoff site: `8k`, site symmetry: `m'.2m'`

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	`[1,7,30,32]`
2	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[2,5,27,28]`

continued ...

Table 11

No.	position	mapping
3	$\left[x,\ -x,\ \frac{3}{4}\right]$	[3,4,26,29]
4	$\left[-x,\ -x,\ \frac{1}{4}\right]$	[6,8,25,31]
5	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[9,15,22,24]
6	$\left[x,\ -x,\ \frac{1}{4}\right]$	[10,13,19,20]
7	$\left[-x,\ x,\ \frac{1}{4}\right]$	[11,12,18,21]
8	$\left[x,\ x,\ \frac{3}{4}\right]$	[14,16,17,23]

Table 12: Wyckoff site: 8l, site symmetry: m2m.

No.	position	mapping
1	$[x,\ 0,\ 0]$	[1,4,13,14]
2	$\left[0,\ x,\ \frac{1}{2}\right]$	[2,7,11,16]
3	$\left[0,\ -x,\ \frac{1}{2}\right]$	[3,8,10,15]
4	$[-x,\ 0,\ 0]$	[5,6,9,12]
5	$\left[x,\ 0,\ \frac{1}{2}\right]$	[17,20,29,30]
6	$[0,\ x,\ 0]$	[18,23,27,32]
7	$[0,\ -x,\ 0]$	[19,24,26,31]
8	$\left[-x,\ 0,\ \frac{1}{2}\right]$	[21,22,25,28]

Table 13: Wyckoff site: 8m, site symmetry: m'2'm.

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,13,20,30]
2	$\left[0,\ x,\ \frac{3}{4}\right]$	[2,16,23,27]
3	$\left[0,\ -x,\ \frac{3}{4}\right]$	[3,15,24,26]
4	$\left[x,\ 0,\ \frac{3}{4}\right]$	[4,14,17,29]
5	$\left[-x,\ 0,\ \frac{3}{4}\right]$	[5,9,22,28]
6	$\left[-x,\ 0,\ \frac{1}{4}\right]$	[6,12,21,25]
7	$\left[0,\ x,\ \frac{1}{4}\right]$	[7,11,18,32]
8	$\left[0,\ -x,\ \frac{1}{4}\right]$	[8,10,19,31]

Table 14: Wyckoff site: 8n, site symmetry: m2m.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ 0\right]$	[1,4,13,14]
2	$\left[\frac{1}{2},\ x,\ \frac{1}{2}\right]$	[2,7,11,16]
3	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[3,8,10,15]
4	$\left[-x,\ \frac{1}{2},\ 0\right]$	[5,6,9,12]
5	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[17,20,29,30]
6	$\left[\frac{1}{2},\ x,\ 0\right]$	[18,23,27,32]
7	$\left[\frac{1}{2},\ -x,\ 0\right]$	[19,24,26,31]
8	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	[21,22,25,28]

Table 15: Wyckoff site: `8o`, site symmetry: `m'2'm`.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,13,20,30]`
2	$\left[\frac{1}{2},\ x,\ \frac{3}{4}\right]$	`[2,16,23,27]`
3	$\left[\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	`[3,15,24,26]`
4	$\left[x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,14,17,29]`
5	$\left[-x,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,9,22,28]`
6	$\left[-x,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[6,12,21,25]`
7	$\left[\frac{1}{2},\ x,\ \frac{1}{4}\right]$	`[7,11,18,32]`
8	$\left[\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	`[8,10,19,31]`

Table 16: Wyckoff site: `16p`, site symmetry: `m..`

No.	position	mapping
1	$[x,\ y,\ 0]$	`[1,14]`
2	$\left[-y,\ x,\ \frac{1}{2}\right]$	`[2,11]`
3	$\left[y,\ -x,\ \frac{1}{2}\right]$	`[3,10]`
4	$[x,\ -y,\ 0]$	`[4,13]`
5	$[-x,\ y,\ 0]$	`[5,12]`
6	$[-x,\ -y,\ 0]$	`[6,9]`
7	$\left[y,\ x,\ \frac{1}{2}\right]$	`[7,16]`
8	$\left[-y,\ -x,\ \frac{1}{2}\right]$	`[8,15]`
9	$\left[x,\ y,\ \frac{1}{2}\right]$	`[17,30]`
10	$[-y,\ x,\ 0]$	`[18,27]`
11	$[y,\ -x,\ 0]$	`[19,26]`
12	$\left[x,\ -y,\ \frac{1}{2}\right]$	`[20,29]`
13	$\left[-x,\ y,\ \frac{1}{2}\right]$	`[21,28]`
14	$\left[-x,\ -y,\ \frac{1}{2}\right]$	`[22,25]`
15	$[y,\ x,\ 0]$	`[23,32]`
16	$[-y,\ -x,\ 0]$	`[24,31]`

Table 17: Wyckoff site: `16q`, site symmetry: `m'..`

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	`[1,30]`
2	$\left[-y,\ x,\ \frac{3}{4}\right]$	`[2,27]`
3	$\left[y,\ -x,\ \frac{3}{4}\right]$	`[3,26]`
4	$\left[x,\ -y,\ \frac{3}{4}\right]$	`[4,29]`
5	$\left[-x,\ y,\ \frac{3}{4}\right]$	`[5,28]`
6	$\left[-x,\ -y,\ \frac{1}{4}\right]$	`[6,25]`
7	$\left[y,\ x,\ \frac{1}{4}\right]$	`[7,32]`
8	$\left[-y,\ -x,\ \frac{1}{4}\right]$	`[8,31]`
9	$\left[-x,\ -y,\ \frac{3}{4}\right]$	`[9,22]`
10	$\left[y,\ -x,\ \frac{1}{4}\right]$	`[10,19]`
11	$\left[-y,\ x,\ \frac{1}{4}\right]$	`[11,18]`
12	$\left[-x,\ y,\ \frac{1}{4}\right]$	`[12,21]`

continued ...

Table 17

No.	position	mapping
13	$\left[x,\, -y,\, \frac{1}{4}\right]$	[13,20]
14	$\left[x,\, y,\, \frac{3}{4}\right]$	[14,17]
15	$\left[-y,\, -x,\, \frac{3}{4}\right]$	[15,24]
16	$\left[y,\, x,\, \frac{3}{4}\right]$	[16,23]

Table 18: Wyckoff site: 16r, site symmetry: ..m'

No.	position	mapping
1	$[x,\, x,\, z]$	[1,32]
2	$\left[-x,\, x,\, z+\frac{1}{2}\right]$	[2,28]
3	$\left[x,\, -x,\, z+\frac{1}{2}\right]$	[3,29]
4	$[x,\, -x,\, -z]$	[4,26]
5	$[-x,\, x,\, -z]$	[5,27]
6	$[-x,\, -x,\, z]$	[6,31]
7	$\left[x,\, x,\, \frac{1}{2}-z\right]$	[7,30]
8	$\left[-x,\, -x,\, \frac{1}{2}-z\right]$	[8,25]
9	$[-x,\, -x,\, -z]$	[9,24]
10	$\left[x,\, -x,\, \frac{1}{2}-z\right]$	[10,20]
11	$\left[-x,\, x,\, \frac{1}{2}-z\right]$	[11,21]
12	$[-x,\, x,\, z]$	[12,18]
13	$[x,\, -x,\, z]$	[13,19]
14	$[x,\, x,\, -z]$	[14,23]
15	$\left[-x,\, -x,\, z+\frac{1}{2}\right]$	[15,22]
16	$\left[x,\, x,\, z+\frac{1}{2}\right]$	[16,17]

Table 19: Wyckoff site: 16s, site symmetry: .m.

No.	position	mapping
1	$[x,\, 0,\, z]$	[1,13]
2	$\left[0,\, x,\, z+\frac{1}{2}\right]$	[2,16]
3	$\left[0,\, -x,\, z+\frac{1}{2}\right]$	[3,15]
4	$[x,\, 0,\, -z]$	[4,14]
5	$[-x,\, 0,\, -z]$	[5,9]
6	$[-x,\, 0,\, z]$	[6,12]
7	$\left[0,\, x,\, \frac{1}{2}-z\right]$	[7,11]
8	$\left[0,\, -x,\, \frac{1}{2}-z\right]$	[8,10]
9	$\left[x,\, 0,\, z+\frac{1}{2}\right]$	[17,29]
10	$[0,\, x,\, z]$	[18,32]
11	$[0,\, -x,\, z]$	[19,31]
12	$\left[x,\, 0,\, \frac{1}{2}-z\right]$	[20,30]
13	$\left[-x,\, 0,\, \frac{1}{2}-z\right]$	[21,25]
14	$\left[-x,\, 0,\, z+\frac{1}{2}\right]$	[22,28]
15	$[0,\, x,\, -z]$	[23,27]
16	$[0,\, -x,\, -z]$	[24,26]

Table 20: Wyckoff site: 16t, site symmetry: .m.

No.	position	mapping
1	$\left[x,\ \frac{1}{2},\ z\right]$	[1,13]
2	$\left[\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[2,16]
3	$\left[\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[3,15]
4	$\left[x,\ \frac{1}{2},\ -z\right]$	[4,14]
5	$\left[-x,\ \frac{1}{2},\ -z\right]$	[5,9]
6	$\left[-x,\ \frac{1}{2},\ z\right]$	[6,12]
7	$\left[\frac{1}{2},\ x,\ \frac{1}{2}-z\right]$	[7,11]
8	$\left[\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	[8,10]
9	$\left[x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[17,29]
10	$\left[\frac{1}{2},\ x,\ z\right]$	[18,32]
11	$\left[\frac{1}{2},\ -x,\ z\right]$	[19,31]
12	$\left[x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[20,30]
13	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[21,25]
14	$\left[-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[22,28]
15	$\left[\frac{1}{2},\ x,\ -z\right]$	[23,27]
16	$\left[\frac{1}{2},\ -x,\ -z\right]$	[24,26]

Table 21: Wyckoff site: 32u, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	[2]
3	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	[3]
4	$[x,\ -y,\ -z]$	[4]
5	$[-x,\ y,\ -z]$	[5]
6	$[-x,\ -y,\ z]$	[6]
7	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[7]
8	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[8]
9	$[-x,\ -y,\ -z]$	[9]
10	$\left[y,\ -x,\ \frac{1}{2}-z\right]$	[10]
11	$\left[-y,\ x,\ \frac{1}{2}-z\right]$	[11]
12	$[-x,\ y,\ z]$	[12]
13	$[x,\ -y,\ z]$	[13]
14	$[x,\ y,\ -z]$	[14]
15	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[15]
16	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[16]
17	$\left[x,\ y,\ z+\frac{1}{2}\right]$	[17]
18	$[-y,\ x,\ z]$	[18]
19	$[y,\ -x,\ z]$	[19]
20	$\left[x,\ -y,\ \frac{1}{2}-z\right]$	[20]
21	$\left[-x,\ y,\ \frac{1}{2}-z\right]$	[21]
22	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	[22]
23	$[y,\ x,\ -z]$	[23]
24	$[-y,\ -x,\ -z]$	[24]
25	$\left[-x,\ -y,\ \frac{1}{2}-z\right]$	[25]

continued ...

Table 21

No.	position	mapping
26	$[y,\, -x,\, -z]$	[26]
27	$[-y,\, x,\, -z]$	[27]
28	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[29]
30	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[30]
31	$[-y,\, -x,\, z]$	[31]
32	$[y,\, x,\, z]$	[32]